Deepfake Detection

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Abstract

Deepfake technology has advanced rapidly, making it increasingly difficult to distinguish real videos from manipulated ones. This project explores a practical approach for detecting such manipulated content by focusing on the faces present within video frames. The method begins by extracting a small number of representative frames from each video and using a face detection model to isolate the most prominent face in each frame. These cropped face images are then used to train a computer vision model to recognize subtle patterns that differentiate real videos from deepfakes. The results show that combining simple frame selection, automated face extraction, and a modern image classification model can provide reliable predictions without requiring complex or highly technical processing steps. This work demonstrates that deepfake detection can be approached in a straightforward and effective manner, supporting both future research developments and real-world applications..

Keywords: Deepfake Detection, Face extraction, video forensics, computer vision

1 Introduction

Deepfake technology has rapidly evolved with the rise of powerful generative models, enabling the creation of highly realistic manipulated videos. These deepfakes can replace faces, alter expressions, or mimic identities with remarkable accuracy, making them difficult for the human eye to detect. While these techniques have legitimate uses in entertainment and creative media, they also pose serious threats when used for misinformation, identity fraud, harassment, and political manipulation. As deepfake content becomes increasingly accessible and convincing, the need for reliable detection systems becomes more urgent.

Detecting deepfakes is a challenging task due to the subtle artifacts left by modern synthesis methods. Manipulated videos often appear visually coherent, and inconsistencies may only emerge at the pixel level or through slight deviations in facial dynamics. Traditional video-level forensic techniques struggle to capture these fine-grained irregularities, especially when videos are compressed or degraded. This creates a strong motivation to explore face-focused analysis, as faces are usually the primary target of manipulation and contain the most discriminative cues.

In this project, we adopt a face-centric approach to deepfake detection by isolating and analyzing the key facial regions within video frames. The system extracts evenly spaced frames from each video, applies MTCNN to detect and crop the largest face, and uses these standardized face images to train a lightweight yet effective deep learning classifier. By removing background information and emphasizing facial features, the pipeline enhances the model's ability to detect manipulations that may otherwise remain hidden in full-frame analysis.

The main contributions of this project are as follows. First, we present a complete preprocessing pipeline that converts raw videos into high-quality face crops suitable for training. Second, we integrate EfficientNet-B0 as a classification model, leveraging its strong representation capability while maintaining computational efficiency. Third, we demonstrate a practical workflow that can be applied to large-scale datasets such as FaceForensics++, enabling consistent and interpretable deepfake detection.

2 Results

This section presents the experimental results obtained from the deepfake detection model based on EfficientNet-B0. The performance of the model is evaluated using training and validation metrics, confusion matrix analysis, and output visualizations of the prediction pipeline.

2.1 Training and Validation Loss

To assess how well the model learned over time, the training and validation loss curves were plotted for all epochs. This plot helps observe whether the model is overfitting, underfitting, or learning steadily. A decreasing training loss with a closely aligned validation loss indicates stable learning behavior. This subsection also displays the Loss and accuracy curves for both the training and validation datasets. The plot helps analyze whether the model is improving its classification ability over epochs and whether the generalization performance remains consistent across datasets.

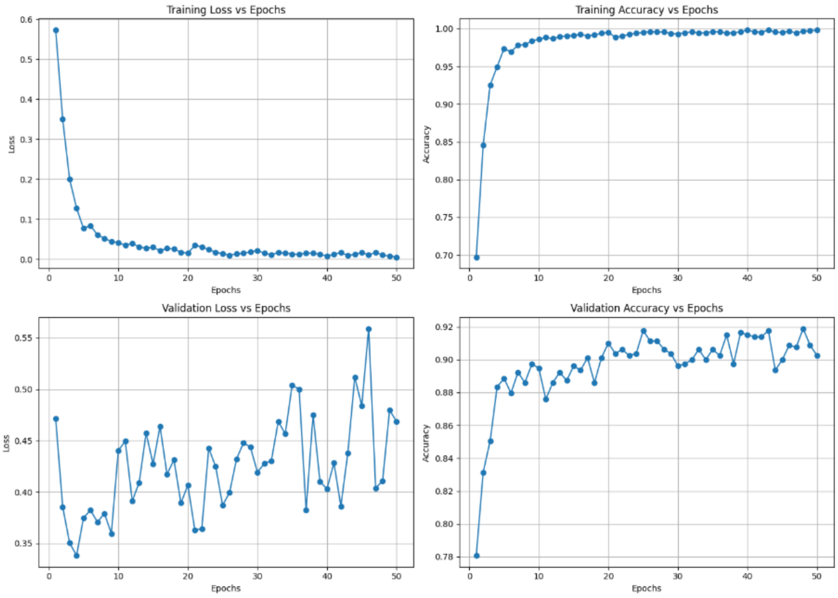

Fig. 1 Training and Validation Loss Curves

2.2 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of the model's predictions on the test/validation set by comparing true labels against predicted labels. This visualization helps identify class-specific strengths and weaknesses of the classifier—such as how well the model distinguishes between real and fake faces.

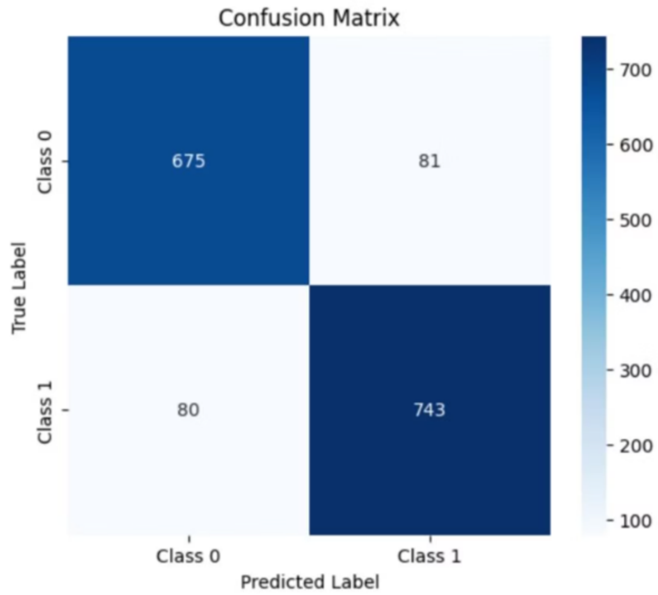

Fig. 2 Confusion Matrix of Deepfake Classifier

2.3 Model Output Visualization

This subsection includes sample output visualizations showing the face detection, bounding boxes, frame-level predictions, confidence scores, and final label for each test video. These results demonstrate how the model behaves in real-world inference scenarios and provide qualitative evidence of prediction reliability.

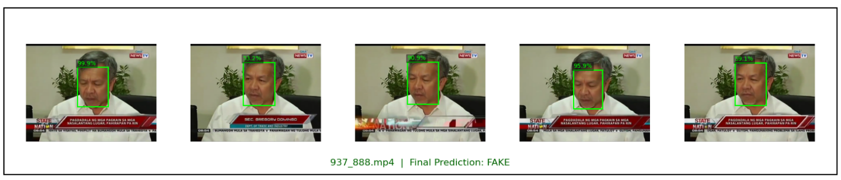

Fig. 3 Sample prediction results from the model (Part 1).

Fig. 4 Sample prediction results from the model (Part 2).

3 Model Comparison and Selection

To determine the most suitable architecture for deepfake detection, multiple convolutional neural network (CNN) models were trained and evaluated on 50% of the dataset. Each model was trained for 20 epochs under identical training settings to ensure a fair comparison. The models were compared based on validation accuracy, stability of training curves, and computational cost. This section summarizes the performance of all evaluated architectures.

3.1 MobileNetV2

MobileNetV2 was selected for experimentation due to its lightweight nature and efficiency on edge devices. However, despite its small size, the model achieved moderate performance with validation accuracy ranging between 79%–80%. While the model trained quickly, the accuracy fluctuations indicate limited feature extraction capability for deepfake detection.

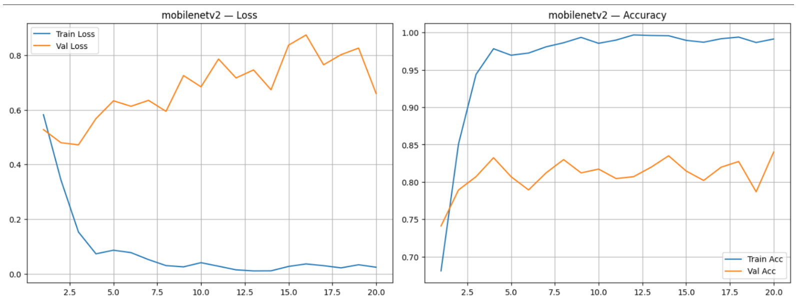

Fig. 5 Training and Validation Loss/Accuracy for MobileNetV2

3.2 MobileNetV3-Large

MobileNetV3-Large also offers low computational cost but showed only slightly improved performance compared to MobileNetV2. With validation accuracy around 80%–81%, the model still struggled to capture deeper visual cues required for identifying face manipulation.

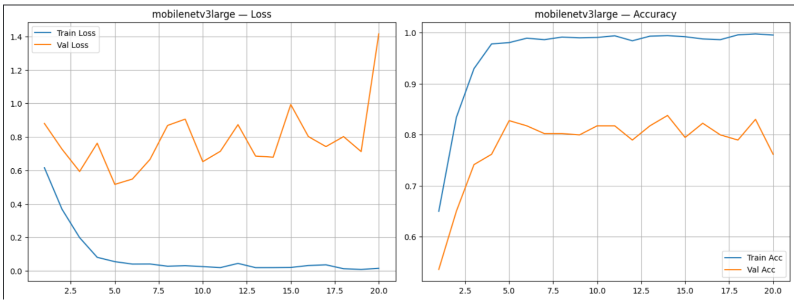

Fig. 6 Training and Validation Loss/Accuracy for MobileNetV3-Large

3.3 EfficientNet-B4

EfficientNet-B4 is a large and computationally heavy model that theoretically offers stronger feature extraction. However, on this dataset, the model underperformed with validation accuracy ranging from 77%–79%. The heavy architecture also resulted in slower convergence, suggesting that the model may be overparameterized for this particular task and dataset size.

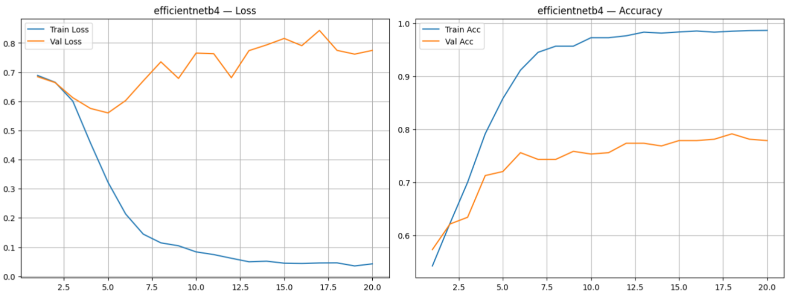

Fig. 7 Training and Validation Loss/Accuracy for EfficientNet-B4

3.4 ResNet50

ResNet50 demonstrated strong performance due to its deeper residual learning blocks. Achieving validation accuracy between 83%–86%, the model consistently showed stable convergence and strong generalization. However, the model is relatively heavy, requiring more computation and memory during both training and inference.

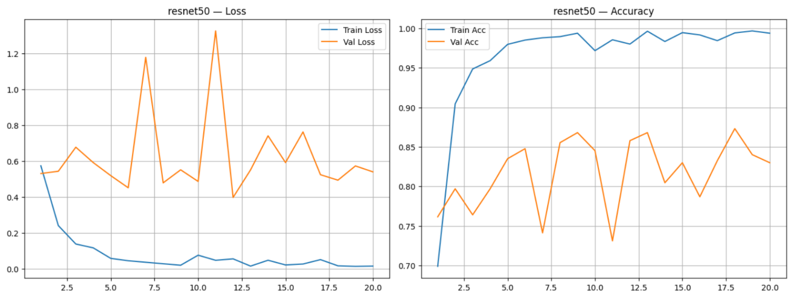

Fig. 8 Training and Validation Loss/Accuracy for ResNet50

3.5 EfficientNet-B0

EfficientNet-B0 achieved strong overall performance while maintaining a small model size. With validation accuracy between 83%–85%, it performed comparably to ResNet50 but with significantly lower computational cost. The model showed stable training curves and strong feature extraction ability relative to its lightweight architecture.

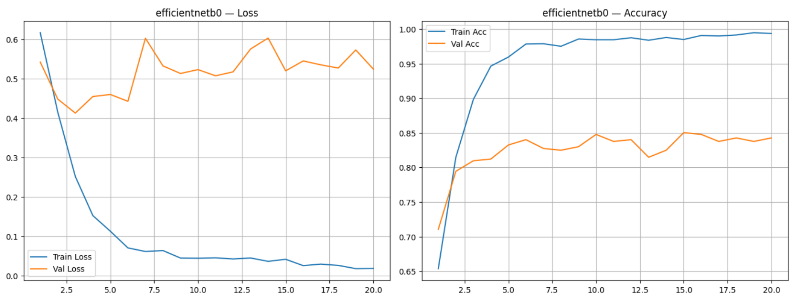

Fig. 9 Training and Validation Loss/Accuracy for EfficientNet-B0

3.6 Final Model Selection

EfficientNet-B0 was selected as the final architecture due to its optimal balance between accuracy and computational efficiency. Although ResNet50 achieved slightly higher peak accuracy, EfficientNet-B0 provided nearly identical performance while requiring fewer parameters and offering faster inference. This makes it more suitable for real-time deepfake detection scenarios where both accuracy and speed are essential.

4 Proposed System

The proposed deepfake detection system focuses on extracting and analyzing facial regions from video data to identify manipulation artifacts. Instead of processing the full video, the system concentrates on the most informative region—the face—where deepfake algorithms typically introduce distortions. The pipeline integrates preprocessing, face detection, frame extraction, deep learning–based classification, and verification to ensure an efficient and accurate detection workflow.

System Pipeline

Figure 10 shows the complete workflow of the proposed system. To insert the image, simply place your file inside the `images` folder and replace the name below.

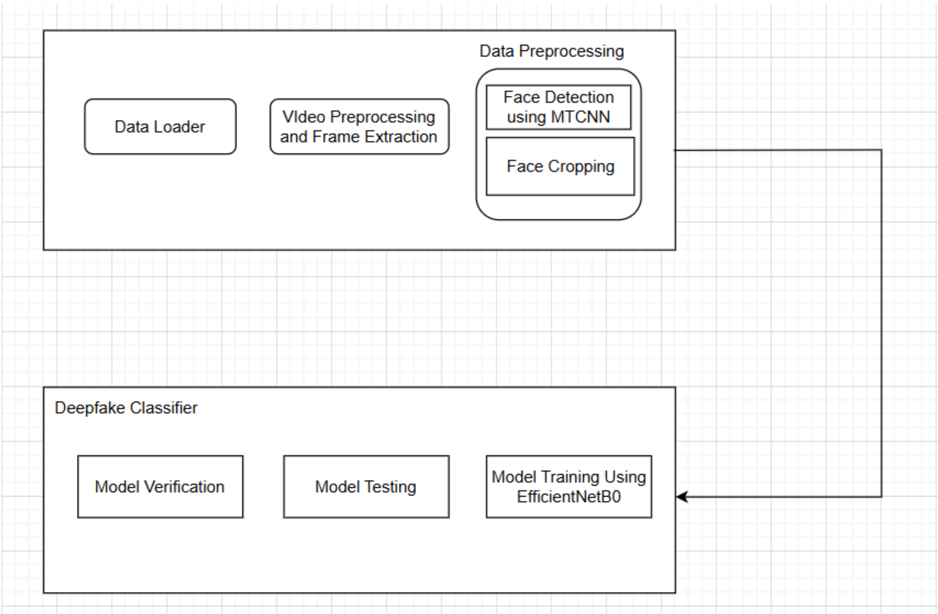

Fig. 10 Overall pipeline of the proposed deepfake detection system.

Model Architecture

The detection model consists of two major components: (1) MTCNN for face detection and cropping (2) EfficientNet-B0 for classification.

Insert your architecture diagram by replacing the filename below.

4.1 Data Loaded

The system uses a curated split of the **FaceForensics++** dataset, one of the largest benchmarks for deepfake research. The dataset used in this project contains:

- **1000 original videos**
- **1000 fake videos** divided into:
  - DeepFakes (200 videos)
  - Face2Face (200 videos)

- FaceSwap (200 videos)
- FaceShifter (200 videos)
- NeuralTextures (200 videos)

Each video contains a single subject speaking in a controlled environment. This ensures consistency and allows the system to focus on facial manipulation artifacts directly.

4.2 Video Preprocessing

Raw videos contain redundant frames, background noise, and lighting variations that do not contribute meaningfully to deepfake detection. Therefore, preprocessing is essential.

The preprocessing pipeline includes:

- **Video decoding** using OpenCV to extract frames.
- **Color conversion** from BGR to RGB for compatibility with MTCNN.
- **Frame standardization** to ensure consistent input resolution.
- **Noise removal** by selecting only relevant frames.

This step ensures that only clean and representative frames are passed to the face detection module.

4.3 Frame Extraction

To represent each video effectively, a fixed number of evenly spaced frames is extracted (e.g., 4 or 5 frames per video). This reduces computational cost while preserving essential temporal changes.

- Frames are selected using a linear spacing strategy.
- Each frame is passed to the MTCNN face detector.
- All detected faces are evaluated, and the **largest face** is selected—assumed to be the main subject.
- The face is **cropped, resized, and normalized** to match the classifier's input requirements.

Add a visual example of extracted and cropped faces:

4.4 Model Training

The processed face images are used to train the **EfficientNet-B0** classifier. This model is chosen for its strong performance-to-efficiency ratio.

Training involves:

- Creating PyTorch datasets and dataloaders.
- Applying normalization and tensor conversion.
- Feeding images into EfficientNet-B0.
- Computing classification loss using cross-entropy.
- Updating weights via backpropagation.

A diagram for the training workflow is shown below. Replace the filename to include your own diagram.

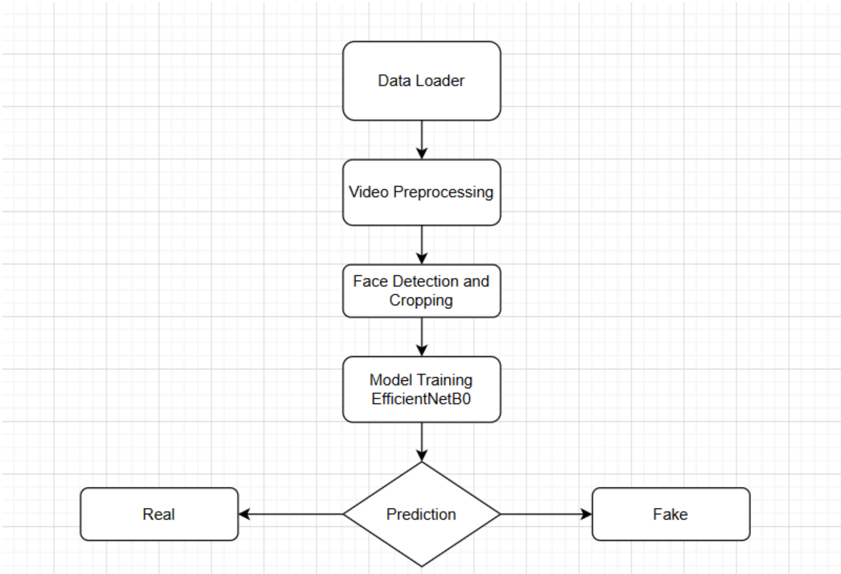

Fig. 11 Training workflow of the deepfake detection model.

4.5 Model Testing

Once trained, the model is evaluated on unseen face images to measure its performance.

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

A testing workflow diagram can be added using the following figure command:

4.6 Model Verification

After testing, the model's predictions are manually verified to ensure reliability.

- Misclassified samples are inspected visually.
- Facial regions are examined for artifacts.
- Differences between predicted and ground-truth labels are analyzed.

This process helps identify limitations and ensures interpretability of the model's decisions.

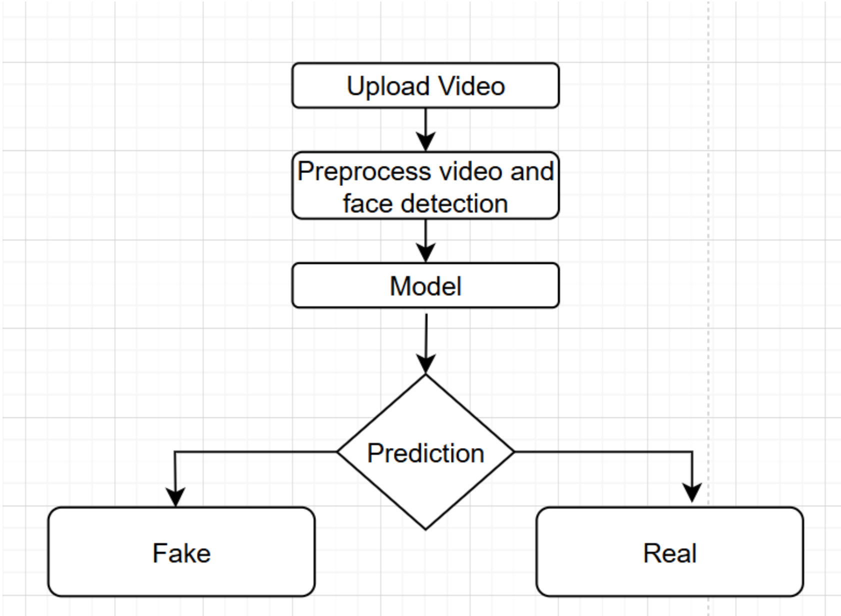

Fig. 12 Testing workflow of the deepfake detection model.

5 Dataset

The proposed system was trained and evaluated using a subset of the **FaceForensics++** dataset. The main characteristics and statistics of the dataset used in this project are summarized below:

- **FaceForensics++** is a large-scale benchmark dataset widely used for research in deepfake generation and detection.
- The curated subset used in this project contains:
  - **1000 original videos**
  - **1000 fake videos**
- The fake videos are evenly distributed across **five manipulation categories**, with:
  - **200 videos per category**
  - DeepFakes (DF)
  - Face2Face (F2F)
  - FaceSwap (FS)
  - FaceShifter (FSh)
  - NeuralTextures (NT)
- Each video contains a single subject speaking in controlled indoor conditions, making the dataset highly suitable for face-focused forensic analysis.
- To create a balanced and computationally manageable dataset, a fixed number of **evenly spaced frames** was extracted from every video.

- The **largest face** in each frame was detected using MTCNN and **cropped** to remove background information and highlight potential manipulation artifacts.
- These cropped facial images formed the final dataset used for training and evaluation, enabling the classifier to learn subtle cues present in manipulated facial regions.

6 Preprocessing

The preprocessing pipeline is designed to extract meaningful facial information from raw video data while eliminating irrelevant background content. The main steps involved are:

- **Frame Extraction:** A fixed number of evenly spaced frames is extracted from each video to ensure consistent sampling and reduce computational load.
- **Face Detection using MTCNN:** Each extracted frame is passed through the MTCNN detector, which identifies all face bounding boxes. The **largest face** is selected to represent the primary subject of the video.
- **Face Cropping:** The selected bounding box is used to crop the facial region from the frame, removing unnecessary background information.
- **Resizing and Standardization:** Each cropped face is resized to a fixed resolution (e.g., 224×224) to match the input requirements of the EfficientNet-B0 classifier.
- **Normalization:** Pixel values are normalized to a standard range to stabilize network training and improve convergence.
- **Tensor Conversion:** All processed images are converted into PyTorch tensors for efficient batching and GPU acceleration.

This structured preprocessing workflow ensures that the deepfake detection model receives high-quality, consistent, and manipulation-focused input data.

7 Tables

Table 1 Classification Report for the Deepfake Detection Model

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Class 0	0.89	0.89	0.89	756
Class 1	0.90	0.90	0.90	823
Accuracy	0.90 (on 1579 samples)			
Macro Avg	0.90	0.90	0.90	1579
Weighted Avg	0.90	0.90	0.90	1579

Table 1 shows balanced performance across both classes, with precision, recall, and F1-scores near 0.90. The model achieves 90% accuracy, demonstrating strong and consistent deepfake detection capability.

8 Conclusion

In this project, we developed a complete and efficient face-centric deepfake detection system by combining frame extraction, face localization using MTCNN, and image classification using EfficientNet-B0. The proposed pipeline focuses specifically on facial regions, which are the primary targets of manipulation in most deepfake generation techniques. By extracting evenly spaced frames from each video and isolating the largest and most informative face, the system ensures that the classifier receives clean and manipulation-focused inputs. This design significantly improves the reliability of the model, even when dealing with compressed or partially corrupted video frames.

The experimental evaluation conducted on a curated subset of the FaceForensics++ dataset demonstrates the effectiveness of the proposed approach. The model achieved an overall accuracy of 90%, along with balanced precision, recall, and F1-scores across real and fake classes. These results highlight the capability of EfficientNet-B0 to capture subtle inconsistencies in manipulated facial regions and validate the suitability of face-focused methods for real-world deepfake detection tasks. Furthermore, the system's modularity makes it adaptable and scalable, enabling future extensions to multi-face scenarios, real-time detection, or integration with more advanced neural architectures.

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